De La Salle University
College of Computer Studies
Department of Software Technology

CSMODEL Major Course Output

(AY 2025-2026, Term 3)

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MCO1 Due Date: June 20, 2026 (Sa) 2130

MCO2 Date: July 25, 2026 (Sa) 2130

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1 Overview

This course project challenges students to identify and evaluate a **real-world dataset** anchored on a research question of their choice, then apply concepts learned in the course to extract meaningful insights. It focuses on exploratory data analysis (EDA) to identify patterns, statistical testing to validate assumptions, and data mining techniques to uncover deeper trends. Students must also interpret results into actionable insights and present them clearly through visualizations and a coherent narrative, strengthening both analytical and communication skills. The project has **two submission phases** with the following deliverables:

- **MCO1**: data description, target research question, preprocessing, and exploratory data analysis
- **MCO2**: statistical inference, data mining, key insights, and conclusions

You are encouraged to work in groups of **at most five (5) members**. Problems with groupmates must be discussed within the group and, if needed, with the instructor.

Each group must ensure that it goes through all of these minimum requirement tasks:

- Identify a research question and/or obtain a dataset, and iteratively explore and refine both; perform exploratory data analysis, covering **at least three (3) EDA questions**, to get a good understanding of the data and ensure alignment between the research question and dataset
- Conduct **at least three (3) statistical tests** to establish three sound conclusions from the data
- **Apply at least one (1) of the following data mining techniques**: (a) rule mining, (b) clustering, or (c) collaborative filtering to discover meaningful insights from the data (you may also choose to apply any of the variants of the above approaches)

Before you start your project, plan what you intend to do with your dataset to ensure it meets the minimum requirements listed above while maintaining the cohesiveness of your data narrative.

2 Deliverables

A **zip file** containing the following:

1. A **single Jupyter notebook** containing all the data processing you did in the project.
   - Submit in both **`.ipynb`** and **PDF** formats.
   - Include the dataset files required to run the project. If the dataset files are too large, include instructions for downloading them.
   - See Section 3 for more information on the structure and content.
2. A **recorded 10-minute video presentation** of you presenting your key findings and insights of your work.
   - Your presentation should be intended for a general audience, using the visual aid listed in the next item.
   - If the video file is too large to submit, you may instead provide a text file containing a link to the video (e.g., YouTube or Google Drive).
   - Refer to Section 4 for more information.
3. A **visual aid** that communicates all key findings and insights of your work.
   - Submit in **PDF** format.
     - For MCO1, a **slide deck** highlighting your initial foundations and EDA findings.
     - For MCO2, a **research poster** summarizing your entire data narrative.
   - Refer to Section 4 for more information.
4. Any other deliverables required by your instructor.

3 Comprehensive Notebook Structure and Content

3.1 Structural Constraints

To maintain readability and analytical focus, the Jupyter notebook must include brief descriptions for each code block that explain the purpose, procedure, and interpretation of results, while avoiding excessive detail that may reduce clarity. To promote concise and focused communication of ideas, the following limits will be imposed:

- Maximum of 5,000 words across all Markdown cells in the final notebook
- Maximum of 100 words per Markdown cell used to describe or explain a code block
- Maximum 15 figures

A Python script will be used to check compliance with these restrictions. **Projects that violate these restrictions will not be checked and will be marked 0 for that MCO phase**.

3.2 Outline and Content

The notebook should be structured in a way that is (a) easy to understand, and (b) can be run sequentially to reproduce all outputs in your work. For the MCO1 deadline, you must submit a **preliminary draft of your notebook** showing your current progress and initial findings. For the MCO2 deadline, you must submit **the final notebook**. Below is the recommended outline of the notebook, with each section containing markdown cells that explain the process and findings.

1 Abstract

This section contains **a high-level overview** of the research question, the dataset used, and the key findings and insights.

2 Dataset Description

This section details the dataset source, collection method, and structure. The following information should be stated in a concise manner:

- A clear, concise description of the dataset
- How the dataset was collected
- Potential implications of how the data was collected on the insights that will be generated
- Structure of the data
    - What each row and column represent (if tabular data)
    - Number of observations
    - What attributes or features are present in each observation
- Brief description of the different attributes in the dataset (if structured), or brief description of the nature of each observation in the data (if unstructured)

Each group must acquire a **real-world dataset** of their choosing. Ensure that the dataset you choose is obtained ethically and legally, in accordance with data privacy laws. The dataset should contain enough variables to explore. As a rule of thumb, a good number would be at least 10 relevant variables (e.g., ID is not a relevant variable for analysis; thus, it is not counted).

You may refer to [this document for a list of possible data sources]. Datasets from other sources, aside from the ones listed in the document, may also be used. Your instructor will provide a sign-up sheet to track which datasets each group is working on.

3 Data Cleaning & Preprocessing

This section documents the steps taken during the data cleaning process to ensure data integrity. Note that there is no need to clean all variables. **Clean only the variables utilized in the study**.

For tabular data, this might entail addressing, but is not limited to: multiple representations of the same categorical value, incorrect data types, missing data, duplicate data, inconsistent formatting, handling of outliers, etc. For unstructured data, check for potential noise or unwanted data, or apply a modeling scheme to convert it to a tabular format first.

4 Research Question & Exploratory Data Analysis

This section states the **general research question** and the iterative process used to refine it, supported by **EDA questions**.

Abide by the following guidelines when coming up with the research question:

- The question does not necessarily have to be groundbreaking, but it must be something that the group is genuinely interested in.
- The question **should not be too specific** (i.e., focused on only a few variables); it should warrant an extensive exploration of the dataset. For example, instead of “Is the average sleep duration correlated with poor academic performance?”, consider “What are the factors associated with poor academic performance?” If your question is too specific, you will have difficulty meeting the project's minimum requirements.
- The question should be feasible to answer given the dataset/s that you have (obviously).

Identifying the research question goes hand in hand with EDA. Sometimes, the EDA can lead you to interesting research questions. On the other hand, you have an initial research question, which is then refined or confirmed by the EDA. In this project, you don’t have to strictly do one before the other. You must iterate between the EDA and question formulation until you arrive at a solid angle supported by the data.

The whole process must cover **at least three (3) EDA questions**. Explicitly state the questions in the Notebook. Having more than 3 questions is acceptable, especially if 3 questions are not enough to get a sufficient understanding of your data. Answer each EDA question using **appropriate numerical summaries** (e.g., measures of central tendency, measures of dispersion, correlation) **and visualizations** (e.g., histograms, distribution plots, box plots). The entire process should be supported by detailed textual descriptions of your procedures and findings.

5 Data Mining

This section details the **application of data mining techniques** to find deeper trends in the data.

Your project must involve **the use of at least one (1) of the following data mining techniques**: rule mining, clustering, or collaborative filtering. Variants of the above approaches, even if not discussed in class, are also welcome, provided that you are able to explain them on a theoretical level. You are also welcome to apply other computational techniques beyond the scope of the course, such as machine learning, provided that you have already satisfied the minimum data mining requirement.

Keep in mind the following reminders when presenting this section:

- If needed, apply preprocessing techniques to transform the data into an appropriate representation before performing data mining. This may include binning, log transformation, conversion to one-hot encoding, normalization, standardization, interpolation, truncation, and feature engineering.
- Some algorithms require the values to be scaled. Make sure to consider this before data mining.
- You are encouraged to use the code from your assignments thoughtfully.

6 Statistical Inference

This section details the statistical inference techniques used to validate or disprove patterns found in the EDA or Data Mining phases.

At least three (3) statistical tests must be performed. These tests can be incorporated into the data narrative wherever appropriate. For example, it can be used to test the significance of findings from EDA questions or the validity of insights generated by data mining techniques.

Keep in mind the following reminders when presenting this section:

- Use statistical inference methods discussed in class.
- Properly state both hypotheses (your hypothesis and the null hypothesis)
- Make sure to show the necessary assumptions and requirements of the statistical tests used, and justify why it is appropriate to use that test for your data. For example, some statistical tests require that the data be approximately normally distributed. If normality assumptions are not satisfied, appropriate non-parametric alternatives may be used.
- Discuss any preprocessing steps you did before computing the p-value.
- Explicitly mention the important values of the tests, such as the p-value and the significance level used.

7 Summary, Conclusions, and Recommendations

This section presents a summary of the project's key findings and insights, highlighting their relevance to the research question and the real-world context. All conclusions must be supported by evidence from the data, including results from statistical tests and analyses. It should also include actionable recommendations based on the results, such as suggested decisions, improvements, or next steps derived from the findings.

Keep in mind that you must produce a **cohesive data narrative** about your dataset. That is, the process you follow in the project, including the data mining and statistical inference requirements, must tie together into a meaningful story centered on your general research question. The insights and conclusions at the end of the Notebook must provide a satisfying resolution to the narrative you are telling.

8 References & Disclosures

This section lists all references used in the project and discloses any tools, datasets, or external resources that contributed to the analysis and development of the study. Refer to Section 7 for more information on the disclosure of generative AI tools.

4 Project Presentation

Be prepared to present your project during the MCO1 and MCO2 deadlines in two formats: (1) a presentation for a general audience and (2) a technical demo.

The **general audience presentation** will be a **recorded video using the phase's designated visual aid**. The following criteria outline the expectations for the recorded presentation:

- The video should be about **10 minutes** long. The delivery and content are for a general audience.
- You are **not required to show yourselves on video**.

- **Voice-overs should be clear**; while we are not requiring high-end audio quality, please ensure the voices are audible and easy to understand. Providing subtitles is optional, but it would be a good idea. Not everyone needs to talk, but everyone must be acknowledged in the video introduction.
- The video should be in **landscape format**; the resolution is up to the group to decide, but it must be of decent quality so that the text on screen is easily readable.

The **visual aid** in the video should convey all key findings and insights from your work.
- For **MCO1**, the recommended content of **slide decks** includes, but is not limited to:
  - A brief explanation of your dataset
  - Your initial research ideas
  - Your exploratory data analysis and research question
  - The project's relevance
- For **MCO2**, you must design a **research poster** summarizing your entire data narrative.
  - The poster must be designed to fit a **standard conference poster size of 48" x 36" in landscape orientation**.
    - You may refer to this Canva template as a guide, but you are free to use any software or tools to create the poster.
    - Ensure that when printed, the poster is readable, even though you are only required to submit a digital copy.
  - The recommended content includes, but is not limited to:
    - Group information
    - Your research question and dataset profile
    - Key findings or insights during EDA, statistical testing, and data mining
    - Conclusions and recommendations

The **technical demo** schedule will be provided by your instructor. Note that **failure to complete the technical demo will result in a score of 0 for that MCO phase**.

5 Grading

For grading, refer to Appendix A. Points are based on the notebook submission, presentation, and answers during the technical demo.

6 Academic Honesty Policy

Honesty policy applies. Please take note that you are NOT allowed to borrow and/or copy-and-paste – in full or in part – any existing related program code or solutions from the internet or other sources (such as printed materials like books, or source codes by other people that are not online). You should develop your own codes and solutions from scratch by yourselves.

The student handbook states that (Sec. 5.2.4.2):

"Faculty members have the right to demand the presentation of a student's ID, to give a grade of 0.0, and to deny admission to class of any student caught cheating under Sec. 5.3.1.1 to Sec. 5.3.1.1.6. The student should immediately be informed of his/her grade and barred from further attending his/her classes."

The student handbook also states that (Sec. 10.3):

A student caught cheating, as defined in Sec. 5.3.1.1., shall be penalized with a grade of 0.0 in the requirement or in the course, at the discretion of the faculty member, without prejudice to an administrative sanction. In cases of alleged cheating, the faculty member should report the incident to the Student Discipline Formation Office (SDFO).

7 Generative AI Use Policy

Following the generative AI use policy stated in the syllabus (allowed in certain contexts), generative AI tools may be used to assist the group in completing the project. However, AI tools should only be used to generate code snippets and should not be used to generate the whole code. More specifically, the majority of the code (at least 80%) should still be written by the students. Non-code blocks, i.e., text blocks, should be entirely written by the students without the help of any AI tool.

Any generative AI use must be accompanied by the following:
- Critical evaluation and understanding of the outputs produced by generative AI before they are incorporated into the project (if you can't explain it, or it doesn't make sense, don't include it).
- Accountability for the resulting output (if the output is wrong, you are the one responsible for it, not AI)
- Disclosure of generative AI use, following the format below.

```
Statement: During the preparation of this work, the author(s) used [NAME TOOL/SERVICE] for the following purposes:

[enumerate a description of all uses of generative AI]

After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.
```

Failing to abide by the above three points when using generative AI will result shall be treated as academic dishonesty and will have grave consequences.

Acknowledgments / Disclosure

During the preparation of this document, the teaching team used the following generative AI tools and/or services:

- ChatGPT was mainly used to streamline and/or construct sentences per section.
- Gemini was used to restructure the outline of this document based on a previous version.

All AI-assisted outputs used in developing this document were carefully reviewed, synthesized, and refined by the teaching team to ensure consistency, accuracy, and alignment with our intended voice and learning outcomes. Following the use of these generative AI tools and/or services in drafting and structuring, the final content reflects the teaching team's own judgment, expertise, and responsibility.

Appendix A: Rubrics

MCO1 Rubrics

Criteria	Ratings				Points
Understanding of Data Collection Method	**PROFICIENT** **6 pts** Shows a good understanding of the data collection process, including its implications for the conclusions.	**SATISFACTORY** **4 pt** Shows a good understanding of the data collection process, but some minor details are missed, or some implications of that process (e.g., biases) are not properly acknowledged in the conclusions.	**DEVELOPING** **2 pt** Shows only a partial understanding of the data collection process.	**NO MARKS** **0 pts** Shows a substantial ignorance of how the data was collected.	6
Understanding of the Dataset	**PROFICIENT** **6 pts** Shows a good understanding of the structure of the dataset, including what each variable or observation represents.	**SATISFACTORY** **4 pt** Shows a good understanding of the structure of the dataset, but some minor details are missed.	**DEVELOPING** **2 pts** Shows only a partial understanding of the structure, variables, and observations of the dataset.	**NO MARKS** **0 pts** Shows a substantial ignorance of the structure or content of the dataset.	6
Justification of Data Cleaning and Preprocessing	**PROFICIENT** **6 pts** Able to demonstrate that sufficient efforts were made to ensure that the dataset is clean and preprocessed, and processes used in data cleaning/preprocessing are justified.	**SATISFACTORY** **4 pt** There are a few data cleaning / preprocessing decisions that are not justified well, or there was not enough justification as to why some data cleaning / preprocessing steps were not applied.	**DEVELOPING** **2 pt** Many data cleaning / preprocessing decisions are not justified well, or there was not enough justification as to why many data cleaning / preprocessing steps were not applied.	**NO MARKS** **0 pts** There was a substantial lack of effort to check if the dataset needs preprocessing / data cleaning.	6
Identification and Thought Process of EDA Questions	**PROFICIENT** **9 pts** At least 3 EDA questions are identified, and the identification of EDA questions is motivated by careful thought and genuine curiosity – not just arbitrarily selected.	**SATISFACTORY** **6 pts** At least 3 EDA questions are identified, but the questions seem arbitrarily selected and not motivated by careful thought and genuine curiosity, or seemingly selected for convenience rather than a desire to explore the data.	**DEVELOPING** **3 pts** Less than 3 EDA questions are identified.	**NO MARKS** **0 pts** No EDA questions are identified.	9
Understanding and Soundness of EDA Methodology and Conclusions	**PROFICIENT** **9 pts** Able to answer all identified EDA questions convincingly (even if fewer than 3), and the conclusions are supported with empirical data through the use of appropriate statistical techniques.	**SATISFACTORY** **6 pts** Able to answer all identified EDA questions (even if fewer than 3), but some of the conclusions are only partially justified with the empirical data.	**DEVELOPING** **3 pts** At least one answer/conclusion does not align with the identified EDA question.	**NO MARKS** **0 pts** Unable to answer any EDA question (this is automatically the mark if there are no EDA questions identified)	9

Presentation and Justification of Exploratory Data Analysis Visualizations	**PROFICIENT** **9 pts** Used appropriate visualization techniques in answering the EDA questions, and was able to justify the selection of those techniques.	**SATISFACTORY** **6 pts** Some visualization techniques are not the most appropriate (i.e., wrong type of plot to use), or there is a failure to justify the selection of some visualization techniques.	**DEVELOPING** **3 pts** Some EDA questions do not have visualization. (Note: the specifications require using visualizations for each EDA question, but an exception can be made to this rule if there is a strong justification on why visualizations cannot be provided for a particular EDA question)	**NO MARKS** **0 pts** There are no visualizations (this is automatically the mark if there are no EDA questions identified).	9
Framing of General Research Question/s	**PROFICIENT** **5 pts** Able to justify that the research question/s is answerable from the dataset, and that the research question is insightful/interesting, but not trivial.	**DEVELOPING** **2 pts** Research question/s partially make sense, but are not insightful/interesting, or too trivial	**NO MARKS** **0 pts** There is no research question/s or the research question/s cannot be answered from the information in the dataset.		5
Organization and Presentation of Notebook	**PROFICIENT** **10 pts** Notebook has a cohesive narrative, is organized well, and includes relevant markdown cells to explain the different parts.	**SATISFACTORY** **7 pts** Notebook narrative is incohesive, slightly lacks organization, or slightly lacks explanation in the markdown cells, making it slightly difficult to navigate through and understand.	**DEVELOPING** **4 pts** Notebook is substantially incohesive, lacks organization, or lacks explanation in the markdown cells, making it difficult to navigate through and understand.	**NO MARKS** **0 pts** There was a substantial disregard for readability in the Notebook, or the Notebook is substantially incomplete.	10
Presentation to General Audience	**PROFICIENT** **15 pts** Clear, well-organized, and fully appropriate for a general audience.	**SATISFACTORY** **10 pts** Presentation is mostly clear and generally appropriate for a general audience, with minor gaps.	**DEVELOPING** **5 pts** The presentation shows limited clarity or audience awareness.	**NO MARKS** **0 pts** Presentation does not meet minimum requirements or was not submitted.	15
Total					**75**

MCO2 Rubrics

Criteria	Ratings				Points
Implementation of Data Mining Technique	**PROFICIENT** **10 pts** At least one data mining technique was applied correctly, and the results were sufficiently interpreted in the context of the research question.	**SATISFACTORY** **7 pts** At least one data mining technique was applied correctly, but the results were not sufficiently interpreted in the context of the research question.	**DEVELOPING** **4 pts** At least one implementation of the data mining technique contains flaws, even though some results are presented.	**NO MARKS** **0 pts** The data mining technique was not implemented.	10

Criterion	Rating	Rating	Rating	Rating	Points
Justification for Data Mining Technique	**PROFICIENT** **5 pts** Able to justify the use of the chosen data mining technique with respect to the question being answered.	**DEVELOPING** **2 pts** The justification of the data mining technique in the context of the question being answered is not completely convincing.	**NO MARKS** **0 pts** There was no justification on how the data mining technique can be helpful in answering the question.		5
Implementation of Statistical Tests	**PROFICIENT** **5 pts** At least three statistical tests correctly implemented, regardless of appropriateness in the context.	**DEVELOPING** **2 pts** At least two statistical tests correctly implemented, regardless of appropriateness in the context.	**NO MARKS** **0 pts** Only one or no statistical tests correctly implemented, regardless of appropriateness in the context.		5
Justification of Statistical Tests	**PROFICIENT** **10 pts** Able to justify all statistical tests in terms of the hypothesis being tested, and able to show that the preconditions for those tests are met.	**SATISFACTORY** **7 pts** At least one of the tests are not justified in terms of the hypothesis being testes, or the preconditions for those tests are met.	**DEVELOPING** **4 pts** At least three statistical tests were implemented correctly, but more than one of them lacking in justification on the applicability of those tests given the underlying conditions of the data.	**NO MARKS** **0 pts** Fewer than three statistical tests were implemented correctly.	10
Overall Conclusions	**PROFICIENT** **10 pts** All final conclusions are supported by relevant data and statistical tests, if needed.	**SATISFACTORY** **7 pts** Conclusions are mostly sound, but there are minor misleading claims, or there are a few claims that are not supported by the data.	**DEVELOPING** **4 pts** Most of the conclusions are misleading, or many conclusions are not supported enough by the data and statistical tests.	**NO MARKS** **0 pts** The conclusions do not answer or align with the defined research questions, or the conclusions are substantially wrong.	10
Organization and Presentation of Notebook	**PROFICIENT** **10 pts** Notebook has a cohesive narrative, organized well, and includes relevant markdown cells to explain the different parts.	**SATISFACTORY** **7 pts** Notebook narrative is incohesive, slightly lacks organization, or slightly lacks explanation in the markdown cells, making it slightly difficult to navigate through and understand.	**DEVELOPING** **4 pts** Notebook is substantially incohesive, lacks organization, or lacks explanation in the markdown cells, making it difficult to navigate through and understand.	**NO MARKS** **0 pts** There was a substantial disregard for readability in the Notebook, or the Notebook is substantially incomplete.	10
Presentation to General Audience	**PROFICIENT** **15 pts** Clear, well-organized, and fully appropriate for a general audience.	**SATISFACTORY** **10 pts** Presentation is mostly clear and generally appropriate for a general audience, with minor gaps.	**DEVELOPIG** **5 pts** The presentation shows limited clarity or audience awareness.	**NO MARKS** **0 pts** Presentation does not meet minimum requirements or was not submitted.	15
Total					**65**